

## **Malware analysis and indicator of compromise (IoC) Investigation**

### **1. Project Overview**

Analyzed a suspicious file reported during security incident using VirusTotal and the Pyramid of Pain framework. The objective was to identify indicators of compromise (IoC), evaluate the malicious nature of the file, and understand the adversary's tactics, techniques, and procedures (TTPs)

### **2. Key Activities**

- investigated a phishing-related malware incident
- Analyzed a malicious file using SHA-256 hashing
- Used TotalVirus to identify associated indicators of compromise
- Mapped findings to the Pyramid of Pain model
- Evaluated attacker infrastructure, tools, and behaviors
- Documented threat intelligence findings and incident response observations

### **3. Indicators of Compromise Identified**

- SHA-256 file hash
- Malicious IP address
- Suspicious domain name
- Network artifacts (HTTP requests)
- Command and control (C2) activity
- Malware tooling associated with the threat actor

### **4. Skills demonstrated**

- threat intelligence analysis
- Malware investigation
- Indicator of compromise (IoC) analysis
- VirusTotal
- Pyramid of Pain
- Incident documentation
- Cyber threat analysis
- Security Operations Center (SOC) fundamentals

### **5. Key learning outcome**

This project demonstrated how threat intelligence platforms can be used to identify malicious artifacts and uncover attacker infrastructure. It also highlighted how different indicators vary in value, with higher-level indicators such as TTPs providing more durable defensive intelligence than simple file hashes.